

LINDENBERG, Germany

WMO: 103930

Lat: 52.2086N

Lon: 14.1181E

Elev: 367

StdP: 14.50

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.4	13.3	2.7	6.4	11.2	8.3	8.5	15.5	27.7	43.2	24.3	43.4	5.5	40	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	85.4	66.2	81.9	64.9	78.8	63.6	68.4	81.3	66.7	78.8	65.1	75.9	5.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.1	90.7	71.7	62.3	85.1	70.5	60.7	80.4	69.4	32.8	81.7	31.5	79.4	30.2	75.8	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.6	16.1	7.2	93.0	9.0	3.6	0.7	95.6	-4.6	97.8	-9.6	99.8	-16.2	102.4		
(5)			6.6	72.0	8.9	1.3	0.2	73.0	-5.0	73.7	-9.9	74.5	-16.4	75.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	31.6	31.9	39.7	46.6	56.4	61.3	65.8	65.3	57.9	49.4	39.2	34.2		
	DBStd	14.26	10.21	8.97	6.90	6.77	7.05	6.44	6.13	6.31	5.35	7.15	6.44	7.20		
	HDD50	2485	571	508	323	145	23	1	0	0	2	98	324	490		
	HDD65	6299	1035	927	783	551	277	149	67	76	223	484	773	954		
	CDD50	1894	0	1	5	44	221	338	489	473	240	80	2	1		
	CDD65	235	0	0	0	0	10	37	91	85	11	1	0	0		
	CDH74	2047	0	0	0	3	103	344	807	710	76	4	0	0		
	CDH80	535	0	0	0	0	5	74	229	220	7	0	0	0		
	WSAvg	7.2	9.0	7.8	8.3	6.9	6.1	6.2	6.3	5.9	6.6	7.1	7.8	8.4		
	PrecAvg	21.8	1.6	1.3	1.5	1.4	2.1	2.4	2.6	2.4	1.7	1.6	1.7	1.8		
(15) Precipitation	PrecMax	31.2	4.0	3.3	4.0	3.7	6.7	6.1	7.6	8.4	5.6	5.0	4.3	4.3		
	PrecMin	14.1	0.0	0.0	0.3	0.0	0.4	0.3	0.2	0.5	0.2	0.2	0.0	0.1		
	PrecStd	4.4	0.7	0.7	0.8	0.8	1.3	1.3	1.6	1.5	1.0	1.0	0.8	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	57.7	63.3	72.5	80.2	87.0	89.9	92.8	80.7	72.5	56.7	52.9		
		MCWB	47.5	50.1	51.3	56.4	62.5	66.2	68.5	69.1	64.5	61.2	51.3	48.4		
	2%	DB	48.0	51.4	57.0	66.9	77.0	81.7	85.4	85.5	75.9	66.6	52.9	50.5		
		MCWB	44.7	46.4	48.6	53.7	60.2	64.0	66.2	66.7	62.5	58.1	48.9	47.2		
	5%	DB	45.7	46.5	52.9	63.0	73.6	78.0	82.2	81.3	71.5	62.7	50.6	47.6		
		MCWB	42.9	42.8	45.7	51.7	58.6	62.6	65.1	65.2	60.7	55.6	47.2	44.9		
	10%	DB	43.4	42.8	49.6	58.7	69.6	74.1	78.7	77.4	67.3	59.5	47.9	44.9		
		MCWB	40.8	39.7	44.3	49.5	56.9	61.0	64.1	63.9	58.6	53.7	45.4	42.9		
	0.4%	WB	47.8	50.6	53.0	57.4	65.3	69.1	71.3	70.2	67.0	63.1	52.1	49.2		
		MCDB	50.6	56.0	60.6	69.1	76.7	79.6	84.1	87.0	77.7	70.2	55.1	52.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.6	46.3	49.3	55.0	62.3	66.5	68.4	68.3	63.9	59.3	49.5	47.6		
		MCDB	47.9	50.2	55.3	65.0	72.4	77.9	80.9	82.2	72.3	65.0	52.3	50.3		
	5%	WB	43.1	43.2	47.1	52.5	59.8	64.2	66.7	66.6	61.9	56.6	47.5	45.1		
		MCDB	45.2	46.2	51.6	61.2	70.0	74.0	78.8	78.9	68.9	61.5	50.1	47.3		
	10%	WB	41.0	39.9	44.8	50.2	57.9	62.1	64.9	64.8	60.0	54.4	45.6	42.7		
		MCDB	43.2	42.5	49.3	58.1	68.2	71.3	76.3	75.0	66.3	58.7	47.8	44.9		
	5% DB	MDBR	8.5	10.9	13.0	16.2	17.6	16.7	17.9	17.8	15.3	13.5	9.4	7.6		
		MCDBR	9.4	15.2	18.6	22.1	22.9	23.4	23.9	23.9	21.5	17.9	12.0	9.7		
		MCWBR	8.0	10.7	11.4	12.1	11.1	10.6	10.5	10.7	11.0	10.7	9.4	8.3		
		MCWBR	9.1	14.5	17.0	20.4	20.5	20.0	21.0	21.7	18.8	16.4	11.6	9.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.1	10.6	11.2	11.7	11.0	10.1	10.6	10.4	10.4	10.6	9.6	8.3		
	Clear-Sky Solar Irradiance	taub	0.323	0.350	0.390	0.412	0.410	0.407	0.427	0.421	0.383	0.373	0.348	0.317		
		taud	2.314	2.291	2.199	2.217	2.266	2.300	2.253	2.285	2.367	2.348	2.353	2.332		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	210	238	250	260	267	269	260	255	252	227	198	191		
		Edh at Noon	20	27	36	40	40	39	41	37	31	26	20	17		
(44) All-Sky Solar Radiation	RadStd		212	431	764	1272	1559	1656	1558	1352	980	547	249	156		
			27	76	96	171	183	132	197	121	122	70	36	16		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)		+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page